



NPTEL Course on “Computer System Optimization for ML”

Lecture #7

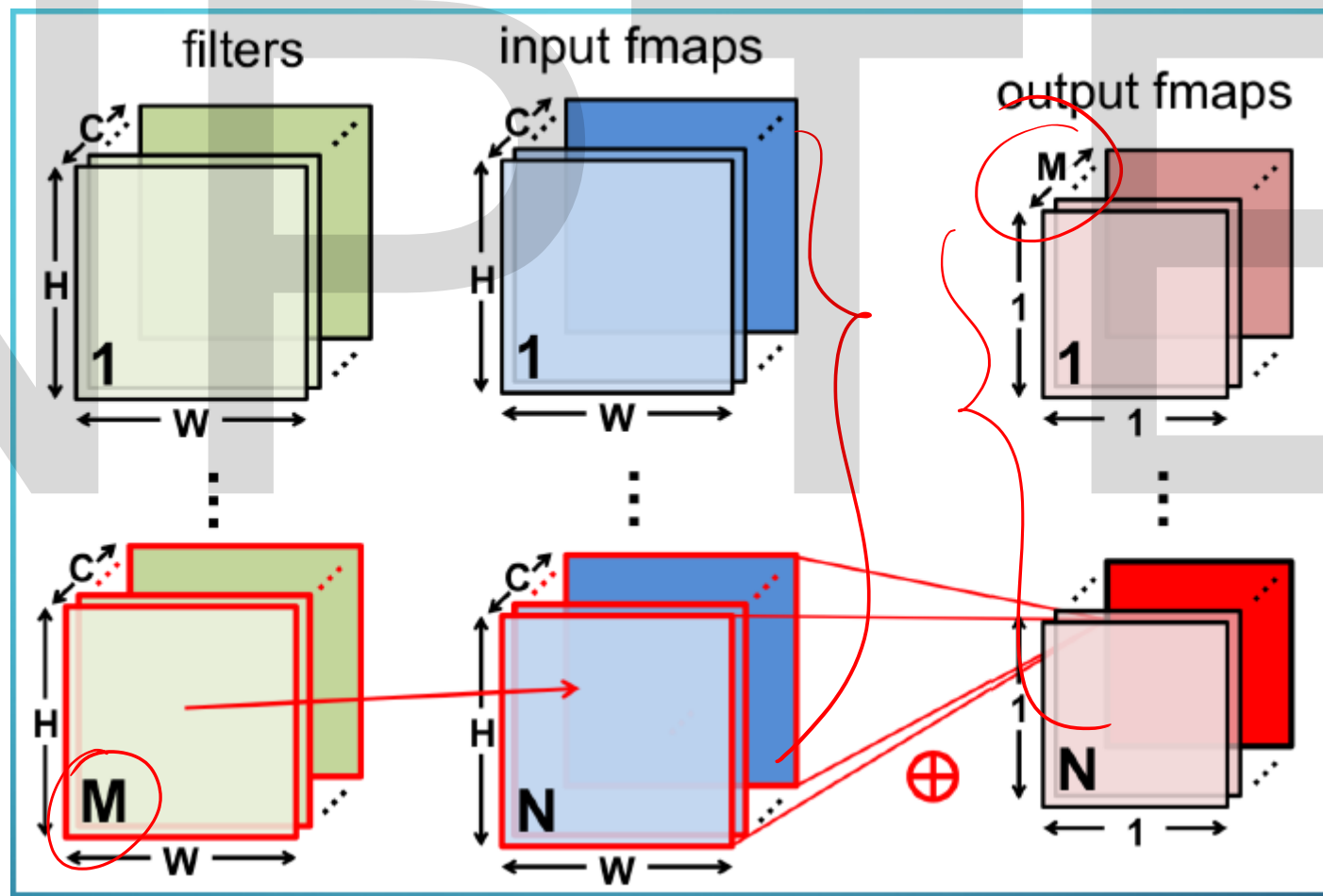


Agenda

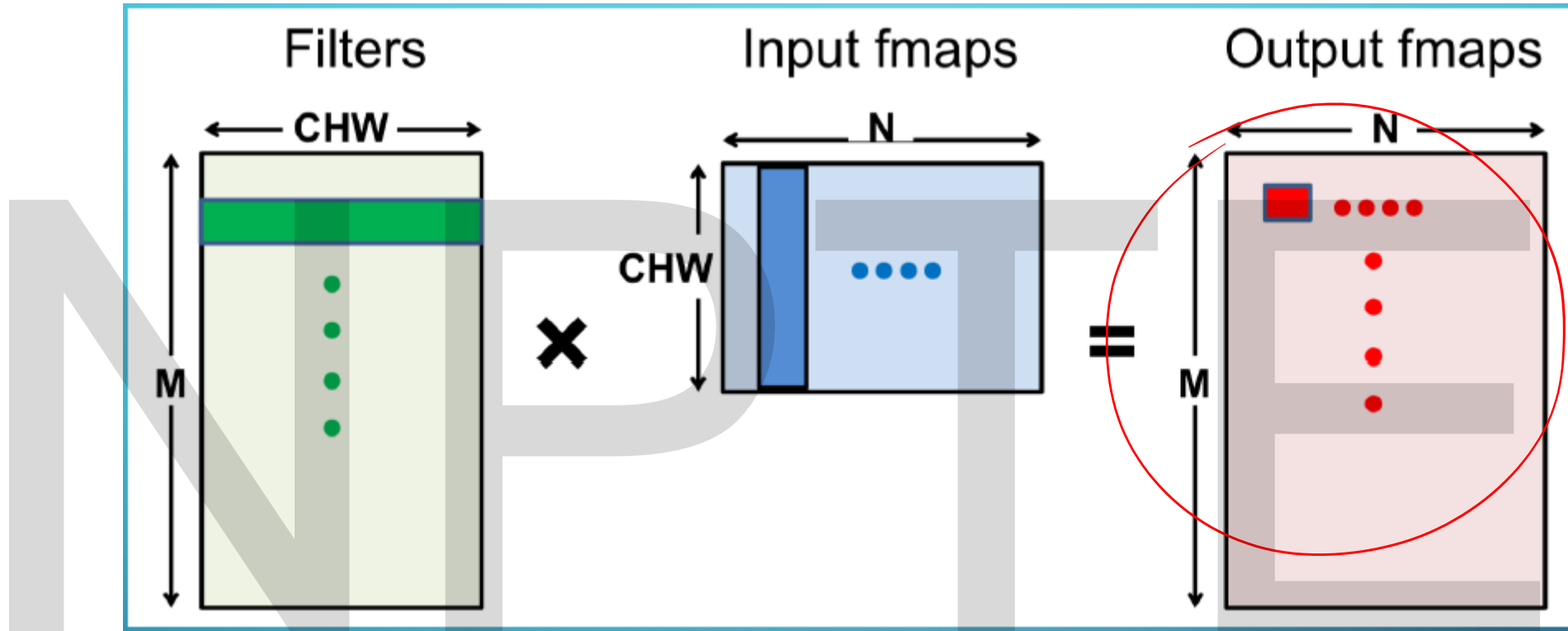
- **Recap**
- **Fully Connected Layer Computation**
- **Non-linear Functions**

Batch of N Input fmaps

- A batch has N input fmaps, so we apply the same M filters of size CHW, and generate N output fmaps of 1x1xM



Flattened Fully Connected Layer



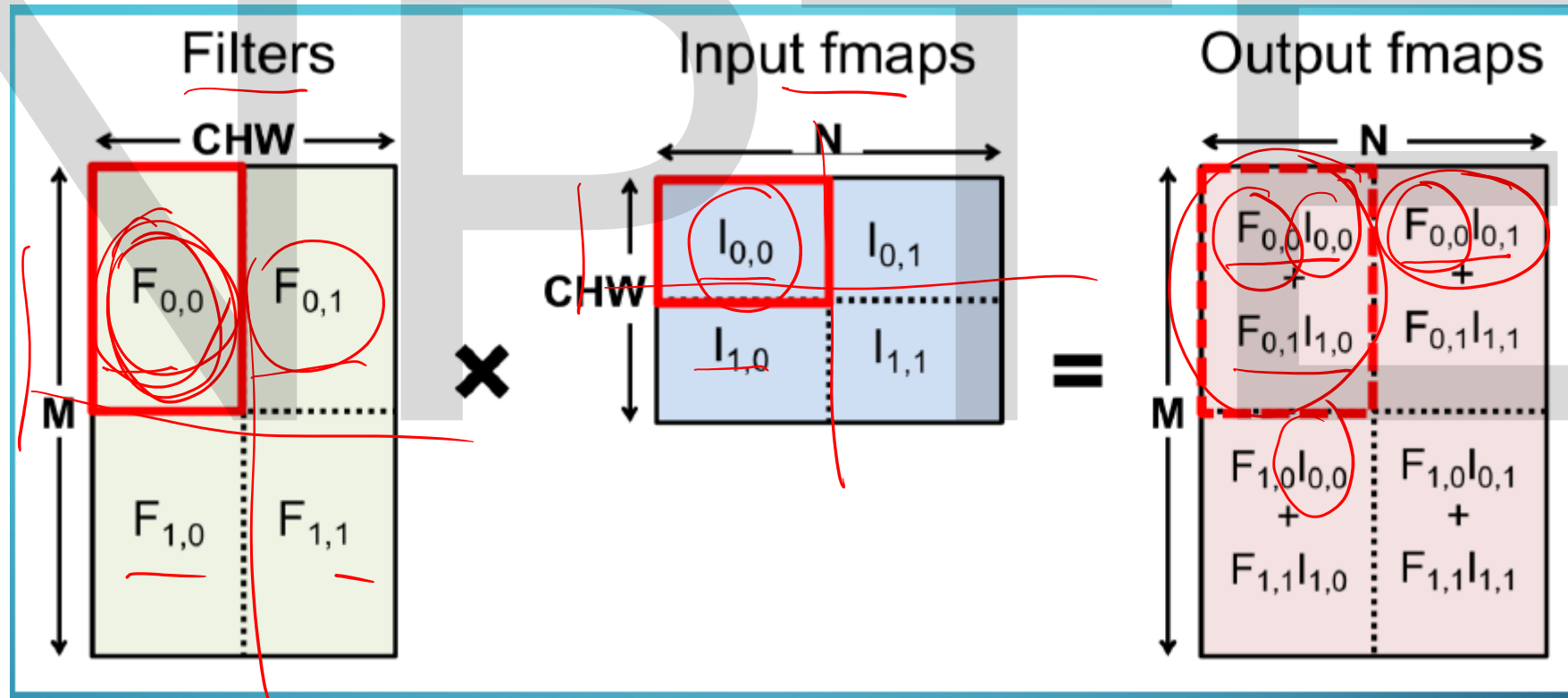
- After flattening, having a batch size of N turns the matrix-vector operation into a matrix-matrix multiplication
- How much temporal locality (reuse of data within a time frame) for this implementation?
 - None

Agenda

- Recap
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Tiled Fully Connected Layer

- Matrix multiplication is tiled to fit in cache
- Computation ordered to maximize reuse of data in cache



Tiled Fully Connected Layer

- **Implementation: Matrix Multiplication (GEMM)**
 - CPU: OpenBLAS, Intel MKL etc
 - GPU: cuBLAS, cuDNN etc
- Library will note shape of the matrix multiplication and select implantation optimized for that shape
- Optimization usually involves proper tiling to storage hierarchy

GV100 – “Tensor Core”

$$D = \begin{pmatrix} A_{0,0} & A_{0,1} & A_{0,2} & A_{0,3} \\ A_{1,0} & A_{1,1} & A_{1,2} & A_{1,3} \\ A_{2,0} & A_{2,1} & A_{2,2} & A_{2,3} \\ A_{3,0} & A_{3,1} & A_{3,2} & A_{3,3} \end{pmatrix} \begin{pmatrix} B_{0,0} & B_{0,1} & B_{0,2} & B_{0,3} \\ B_{1,0} & B_{1,1} & B_{1,2} & B_{1,3} \\ B_{2,0} & B_{2,1} & B_{2,2} & B_{2,3} \\ B_{3,0} & B_{3,1} & B_{3,2} & B_{3,3} \end{pmatrix} + \begin{pmatrix} C_{0,0} & C_{0,1} & C_{0,2} & C_{0,3} \\ C_{1,0} & C_{1,1} & C_{1,2} & C_{1,3} \\ C_{2,0} & C_{2,1} & C_{2,2} & C_{2,3} \\ C_{3,0} & C_{3,1} & C_{3,2} & C_{3,3} \end{pmatrix}$$

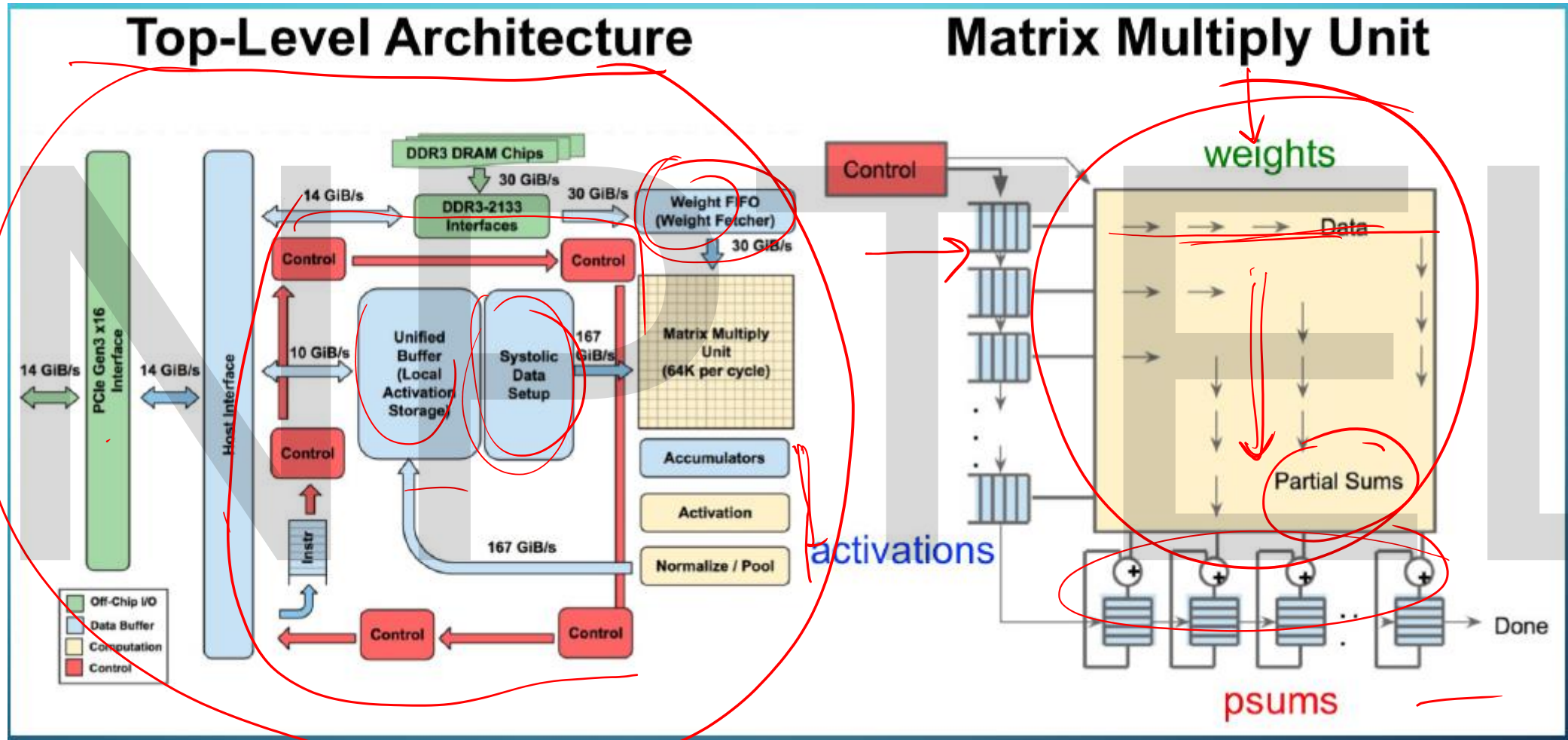
FP16 or FP32 FP16 FP16 FP16 or FP32

- New opcodes
 - Matrix Multiply Accumulate (HMMA)
- FP16 operands
 - 48 inputs / 16 outputs
- 64 multiplies
- 64 adds
- 120 TFLOPS (FP16)
- 400 GFLOPS/W (FP16)

$D = A \times B + C$

4x4

Tensor Processing Unit



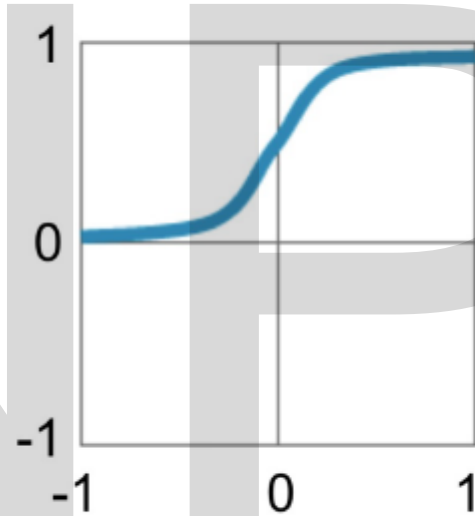
Agenda

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NPTEL

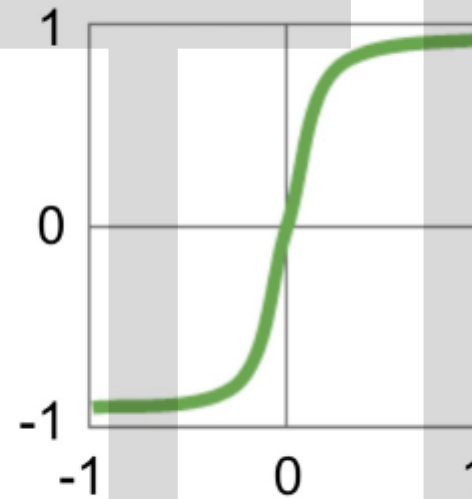
Non-linear Operation

Sigmoid



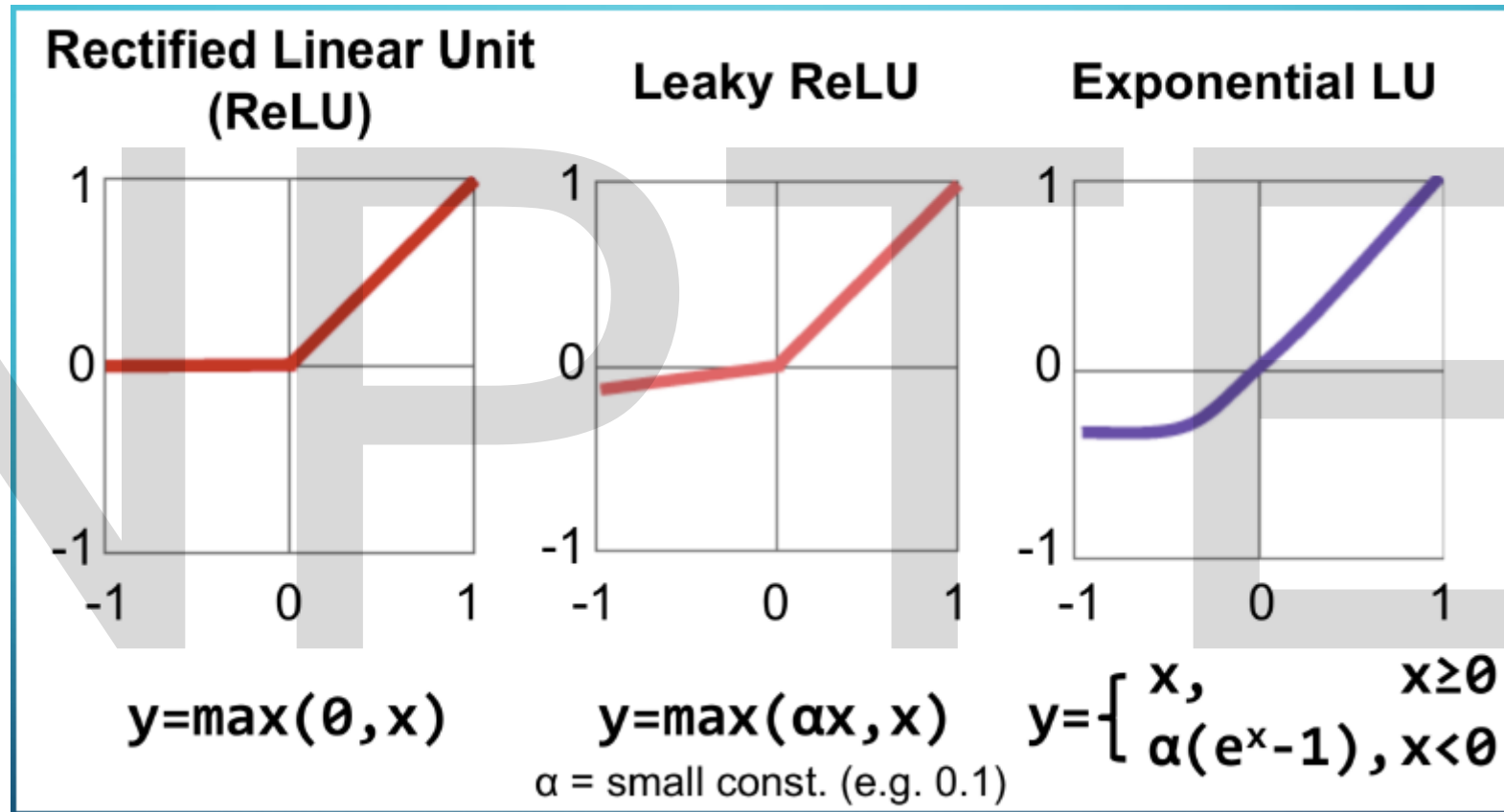
$$y = 1 / (1 + e^{-x})$$

Hyperbolic Tangent



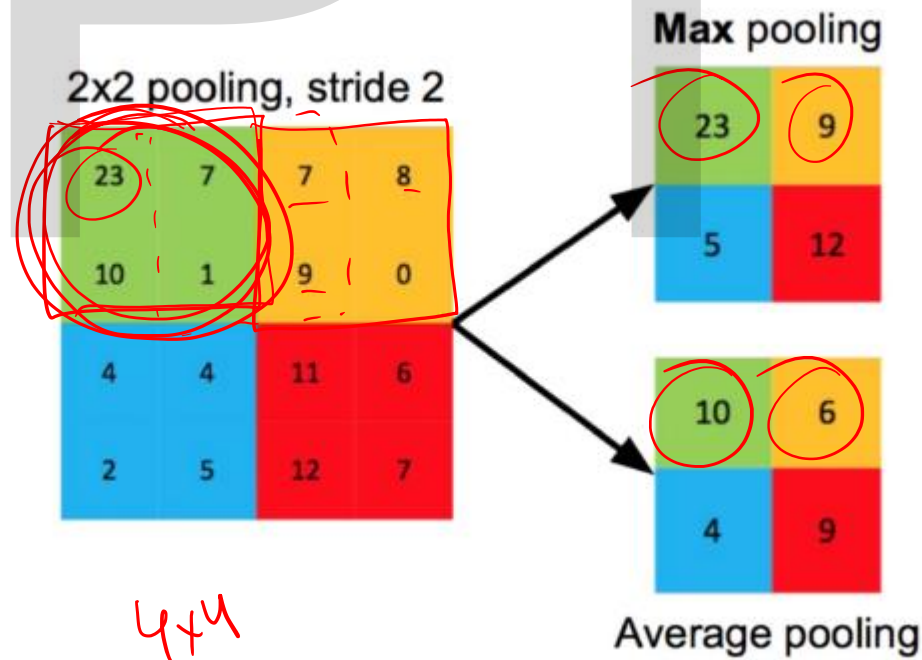
$$y = (e^x - e^{-x}) / (e^x + e^{-x})$$

More Activation Functions

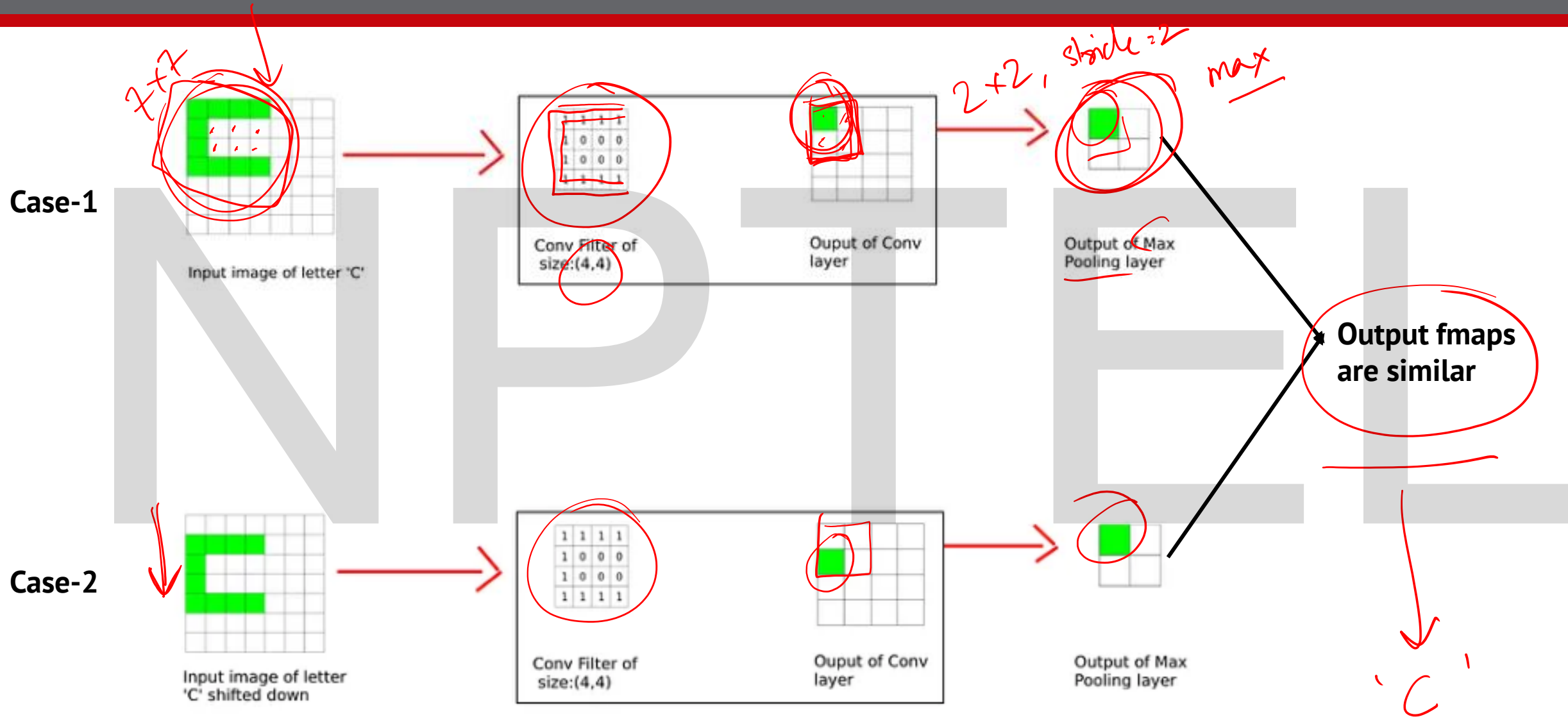


Pooling (Pool) Layer

- Reduce resolution of each channel independently
- Overlapping or non-overlapping
 - Depends on stride
- Increases translational-invariance and noise-resillience

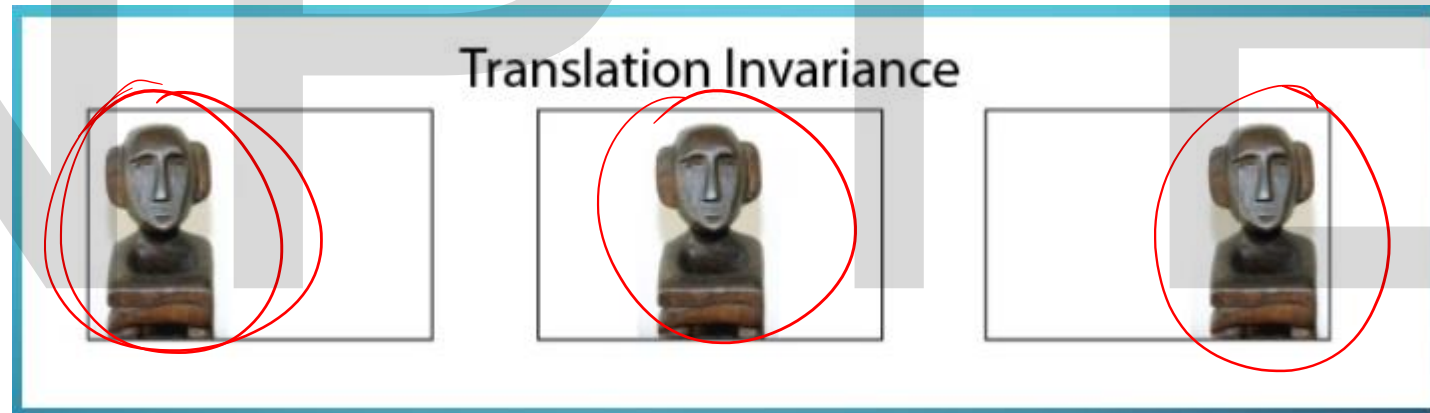


Translational Invariance



Translational Invariance

- Provides the same output independent of the location of the object within the image
- Pooling helps to provide the invariance





THANK YOU

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